

Empirical Analysis for Policymakers

Seminar 1: Introduction and R

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Sciences Po Paris - Reims

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Welcome!

- Who am I?
 - Teaching Fellow in the Department of Economics
 - Research in Behavioral Economics and Applied Microeconometrics
- **Tuesdays.** 16h40-18h40
- Contact: `emirtarik.dakin@sciencespo.fr` or office hours (TBD).

- **Course material:** seminar slides will be uploaded on Moodle, along with readings and other supplementary material.
- This class is a simplified applied microeconometrics course. However, it's not going to be so simple as it will contain a lot of new concepts for the most of you.
 - Try to recap everything we do in class, both theory and practical. Most of these concepts or even your own writing/code will look alien to you after a week if you don't review them.
 - Make sure to have a general understanding of the readings before you come to class and bring notes.
 - Work with each other.
 - Ask questions, clarify doubts as soon as you have them.
- You are **free to use ChatGPT** for coding and understanding concepts. However, consistently using ChatGPT will be obvious in your coding assignments and when you'll be asked questions.
- You will get a **zero** for assignments that use AI text (not code), not to mention a possible disciplinary action for plagiarism. Please be careful.

Assessment and grading

There are **three** assessments for your total grade.

❶ 4 quizzes (%20).

- 20-30min long quiz sessions that will be announced the week before.
- Will be about what we cover in class, mathematical or reasoning.
- Try not to miss these without an excuse.

❷ 3 homework assignments (%30).

- In groups of two.
- R exercises that will require you to be creative or questions about the assigned papers.
- No late submissions allowed.

❸ Empirical project (%50).

- In groups of three.
- You will select a research question along with a data set, review the literature, justify and apply one or more methods we learn in class and answer the research question.
- **Presentations** - November 19th and 26th.
- **Paper deadline** - December 10th.

- This class is a combination of applied topics, starting off with simple regression analysis and moving onto more sophisticated methods for observational data.
- By **observational data**, we mean situations in which the information was collected without manipulation from an investigator. In other words, no experiments involved.
- We'll see much more about this but the main issue with observational data is that sometimes we want to answer research questions that are about **cause and effect**.
- And answering cause and effects questions with observational data is not always straightforward, but much more exciting when you do actually find something.

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- Regression modeling is the most common tool to we use to answer **causal** questions.
- You will have to become good at statistical modeling if you want to pursue academics or research as a career.
- This skill can be acquired by a lot of studying, time, practice and **mastering the basics**.
- The basics includes understanding what **regression model estimates** mean. Might sound simple, but it isn't.
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- This is an **applied** methods class. We won't ignore statistical theory but there is no need to be proving theorems.
- Understanding theory beyond the proofs is the goal of this class. To be a good applied empirical researcher you need to understand different methods and know what theory tells you about them.
- The statistics — and econometrics as eventually follows — revolution of the last 40 years has been the result of raw computing power. This is where R comes into play.

Why regression models?

- There are several reasons to run a regression:
 - **describing** a relationship
 - making **causal** statements and testing hypotheses
 - making **predictions**
- The purpose of running a model changes the way you think about **assumptions and the way you conduct the analysis**. We will often come back to the term **causal inference** when reading papers and analyzing data.
- Anybody can analyze randomized data (run experiments); working with **observational data** is a lot harder.

- Regression models are key to drawing **causal inferences** from observational data.
- We will start by **defining causal effects** and introducing some new language.
- We will review regression results focusing on interpretation and practical issues.
- Then we will talk about **research designs** and their assumptions without using statistical models. You will be thinking about datasets and ways in which you could apply these methods.

Using causal inference

- Does immigration lead to lower wages and/or higher unemployment for locals?
- Does raising the minimum wage reduce employment for low-skilled labor?
- How much effect of an additional year of higher education have on wages?
- Do higher public debt levels lead to lower economic growth?
- Does the neighborhood you grow up in have an effect on your life outcome?

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Research Designs

- By **research design**, we mean the overall strategy used to answer a research question.
- This is different than the **estimation** of effects or the particular regression model or statistical approach we use to answer a research question.
- Traditional econometrics (as in Wooldridge) tends to combine **design** and **estimation**, but that might be more confusing at times.
- For example, whether causality is established or not is discussed in terms of assumptions about the error term $\epsilon \sim N(0, \sigma^2)$ (i.e., **conditional independence assumption**).
- But then in other models like logistic, probit, or Poisson, there is no additive error term and so the intuition of conditional independence gets you tied in knots.
- We'll cover both ways of understanding causal inference. Angrist and Pischke (2009) is a decent bridge (*Mostly Harmless Econometrics*). For a more intuitive approach, we will follow the book *Mastering 'Metrics*, which is also by Angrist and Pischke (2014).

- There is no clean separation in between **design** and **estimation**.
- In some circumstances, our understanding of a research design needs to be adapted to a particular method of estimation.
- Difference-in-differences is an example. Some assumptions hold in the linear world, others can get complicated with nonlinear models.
- **Linear** means a lot of things. Sometimes literally a straight line; sometimes it's figuratively. But it also means **constant**. Sometimes people say linear when they really mean **additive** (as opposed to multiplicative like interactions).

- We will cover the most important and widely used research designs in social sciences research (including economics and political sciences) classified on one main assumption:
 - Selection on observables (ignorability):
 - 1 Regression adjustment
 - 2 Propensity scores and matching
 - No selection on observables (ignorability):
 - 1 Difference-in-differences
 - 2 “Fixed” effects
 - 3 Instrumental variables
 - 4 Regression discontinuity
- **Selection on observables** essentially means whether we observe all confounders or not. If we do, regression adjustment and propensity scores are the right tools. If not, then there is an arsenal of tools with pros and cons. One simple rule of thumb is that with observational data, there is —almost surely— no selection on observables (but could get close enough).

Our statistical tool: R

- This class is also about R.
- R is a means to an end. The end is to manipulate and analyze data.
- There are other tools like Stata, Matlab, SAS, Python. R is the most widely used software and the best place to start. It's the tool of choice for a lot of academics and people in business for several fields; including economics, political sciences, psychology etc.

So let's install R and get started with a couple of projects.

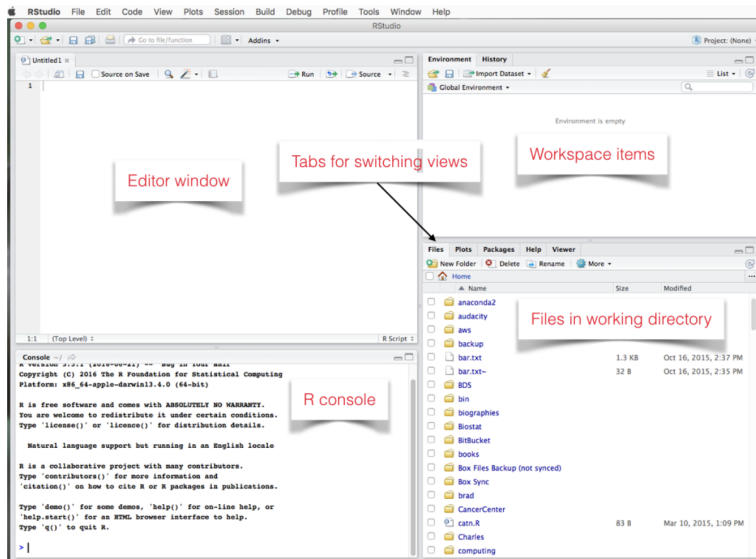
❶ Download R:

- Go to the CRAN website: <https://cran.r-project.org/>
- Click on the link for your operating system (Windows, macOS, Linux).
- Follow the instructions to download and install R.

❷ Download RStudio:

- Visit the RStudio website:
<https://posit.co/download/rstudio-desktop/>
- Click on "Download RStudio" and choose the free version.
- Select your operating system and follow the download and installation instructions.

RStudio Layout



Next...

- More R next week.
- Data wrangling and visualization.
- Exploring Twitter data (not Twitter but actually BlueSky).